

School of Computer Science Engineering and Technology

Lab No. - 24

Total Marks—2

Course-B. Tech.	Type- Core
Course Code- CSET301	Course Name- Artificial Intelligence and Machine Learning
Year- 2025	Semester- Odd
Date-	Batch- 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
Q1	√	√	√			

Lab 21 – Implementation of Vision Transformer (ViT)

Objective

To design and implement a **ViT (ViT)** using **TensorFlow / Keras** for image classification on the **CIFAR-10 dataset**, understanding working of ViT along with positional encoding, and model evaluation metrics.

Problem Statement

Develop and train a ViT model to classify images from the CIFAR-10 dataset into 10 categories (airplane, car, bird, etc.). Build the ViT using TensorFlow, train and evaluate its performance, and visualize predictions.

About Dataset

- **Dataset:** CIFAR-10 (`tensorflow.keras.datasets.cifar10`)
- **Description:** 60,000 color images (32×32×3) in 10 classes, 6,000 images per class.
- **Classes:** airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck.
- **Input:** Image pixels
- **Target:** Class label (0–9)
- **Dataset Source:** [CIFAR-10](#)

Tasks

1. **Data Pre-processing**
 - Load CIFAR-10 dataset.
 - Normalize pixel values to $[0,1]$.
 - Convert labels to one-hot vectors.
 - Split into training and test sets.
2. **ViT Model Design**
 - Patch and Positional Embedding.
 - Transformer Encoder block design
 - Output layer with softmax activation.
3. **Model Compilation**
 - Optimizer: Adam
 - Loss: Categorical Cross-Entropy
 - Metrics: Accuracy
4. **Model Training (Consider following parameters)**
 - Train for 10–20 epochs, batch size 64, validation split 0.2., patch_size=4, embed_dim=192, depth=6, num_heads (self-attention heads)=6, mlp_dim=384, dropout=0.1
5. **Evaluation & Visualization**
 - Evaluate on test data.
 - Plot training/validation accuracy & loss.
 - Confusion matrix & classification report.
 - Visualize some predictions.
6. **Interpretation**
 - Discuss ViT learning patterns, overfitting, and model improvements.